

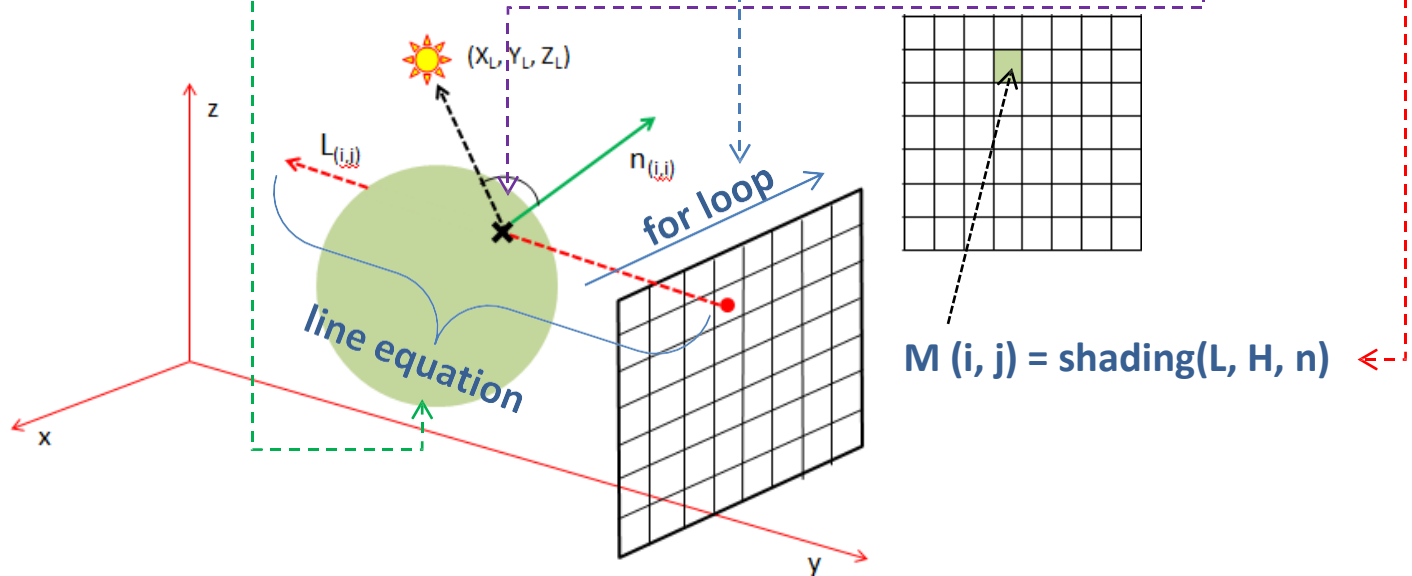
CSE4203: Computer Graphics
Lecture – 9 (part - B)
Ray Tracing

Outline

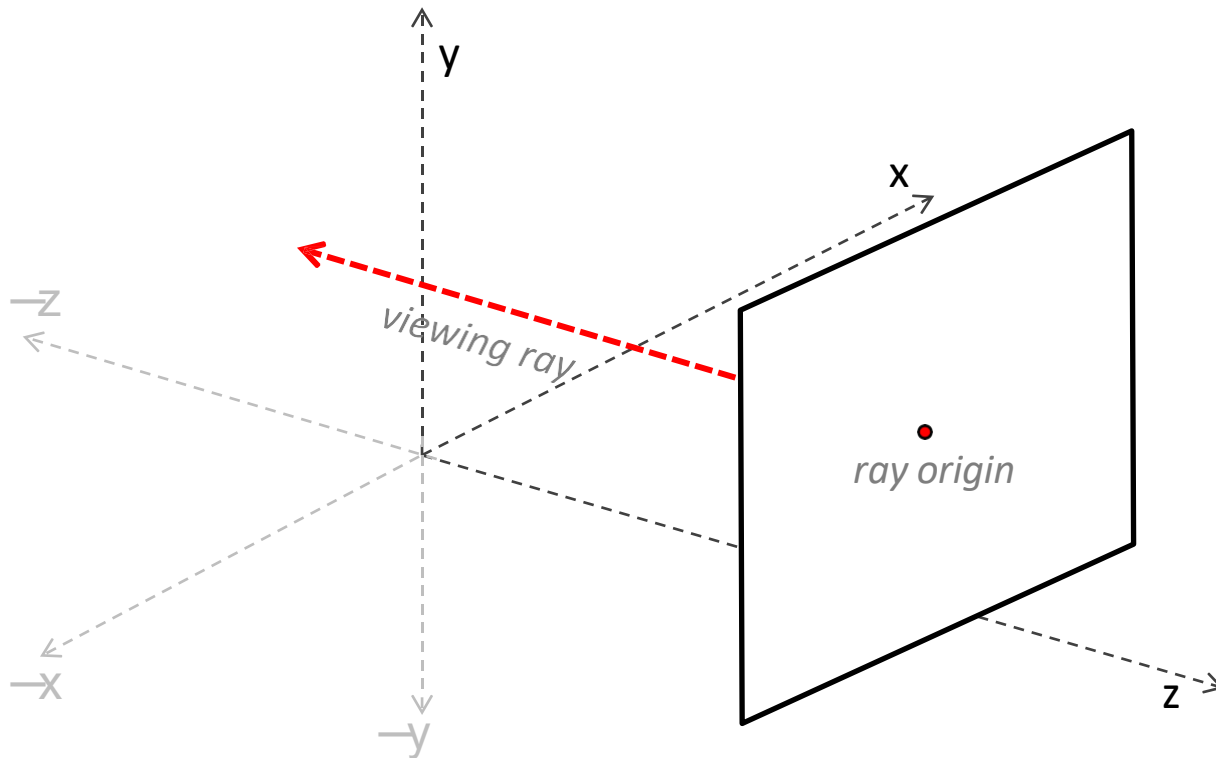
- Ray-tracing
- Camera Frame
- Image Plane and Raster Image
- Computing Viewing Rays
- Ray-sphere Intersection
- Shading

Ray-Tracing Algorithm

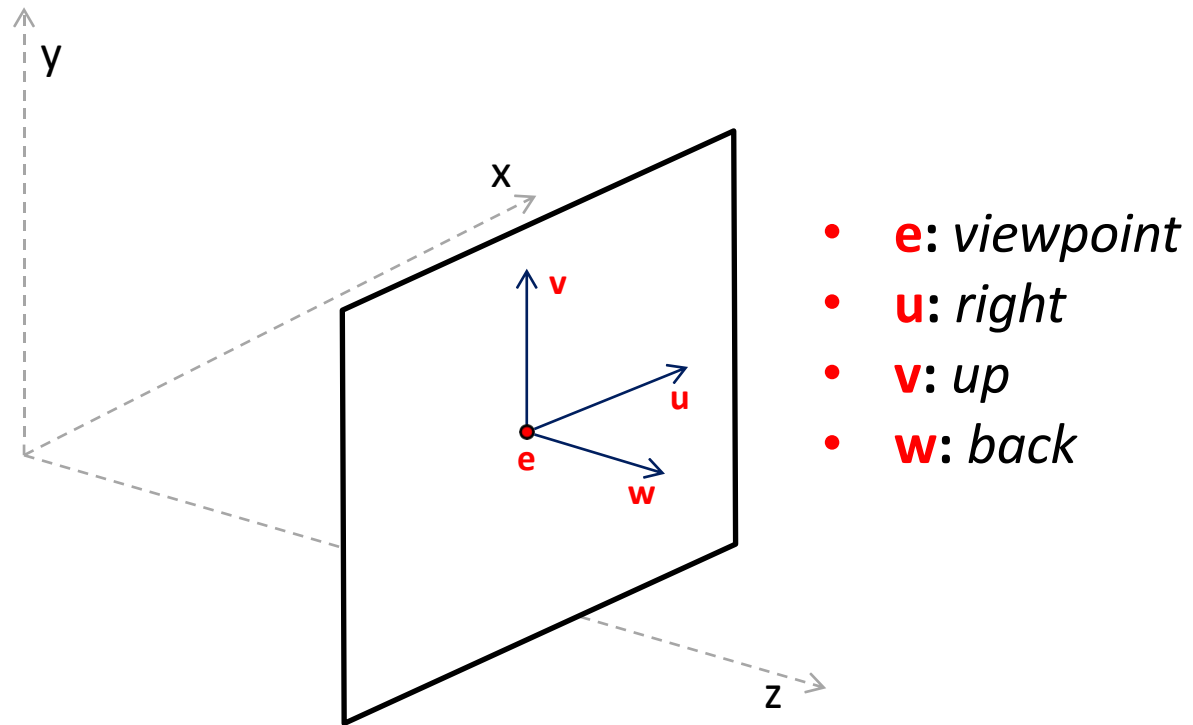
- **for each pixel do:**
 - compute **viewing ray**
 - find first object hit by ray and its **surface normal $\mathbf{n}$**
 - set pixel color computed from **hit point, light, and $\mathbf{n}$**



Camera Frame (1/6)

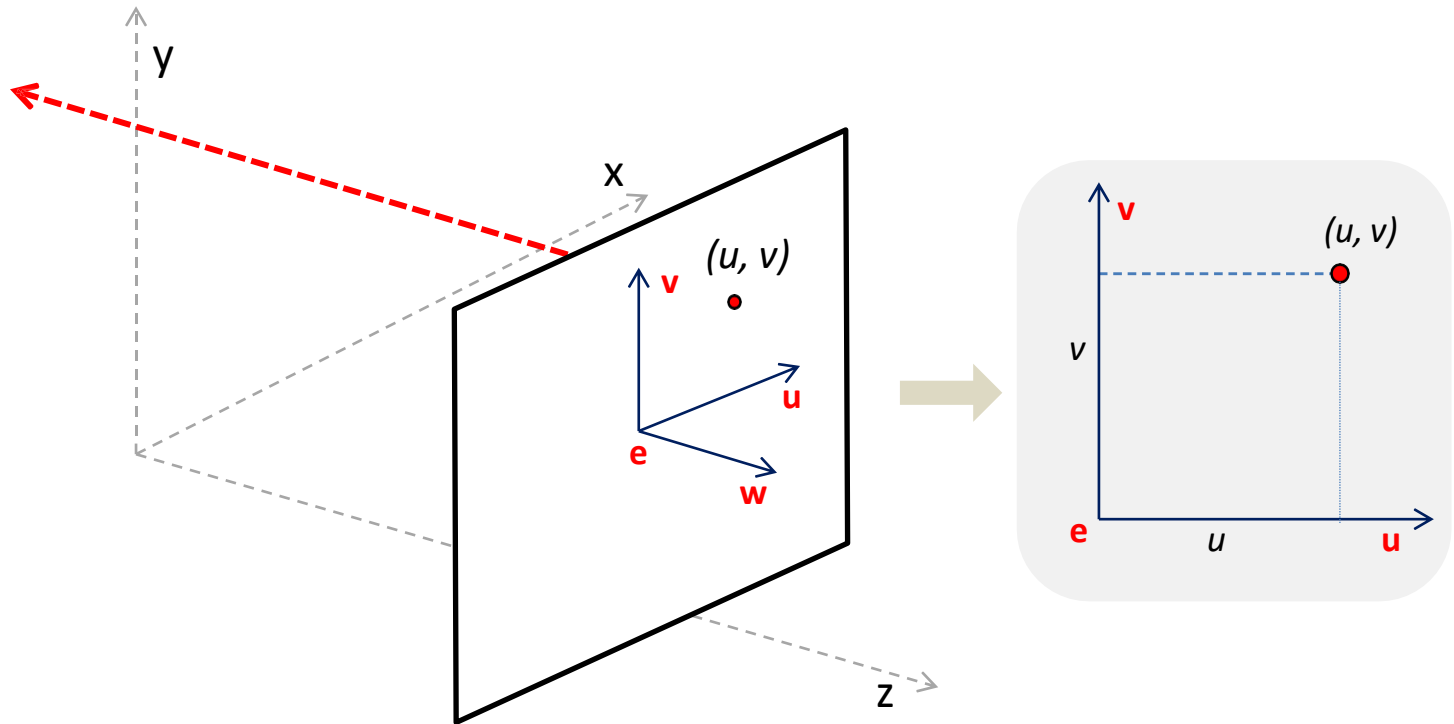


Camera Frame (2/6)

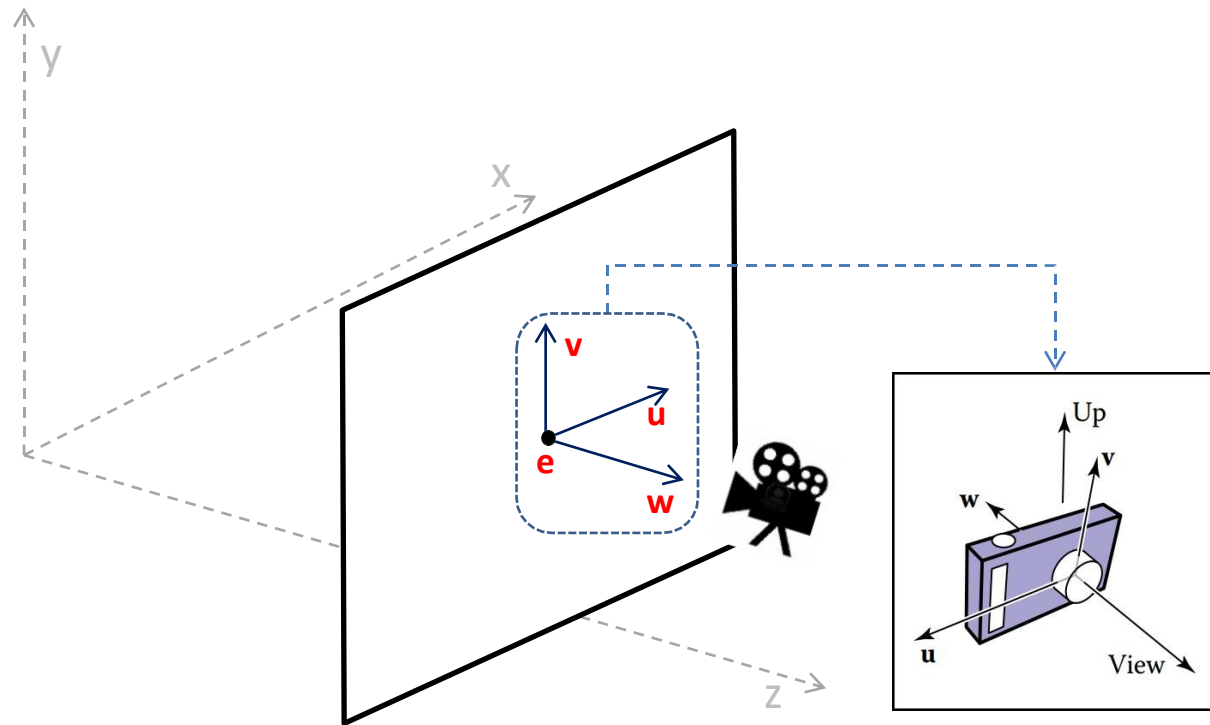


Camera Frame (3/6)

- ray origin = $\mathbf{e} + u \mathbf{u} + v \mathbf{v}$
 - ray direction = $-\mathbf{w}$

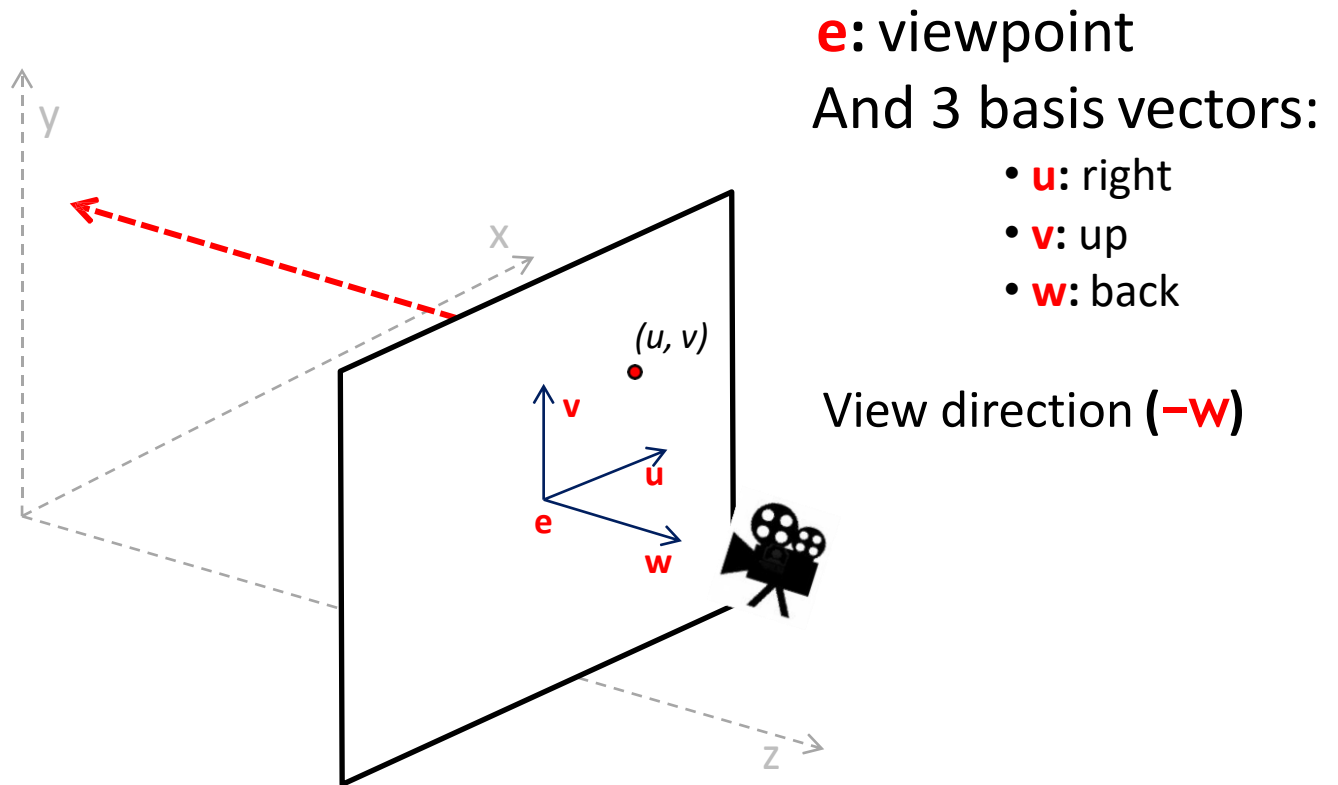


Camera Frame (4/6)



Camera Frame (5/6)

Camera frame: (Camera coordinate)



Camera Frame (6/6)

Orthographic:

- ray direction = $-\mathbf{w}$
- ray origin = $\mathbf{e} + u \mathbf{u} + v \mathbf{v}$

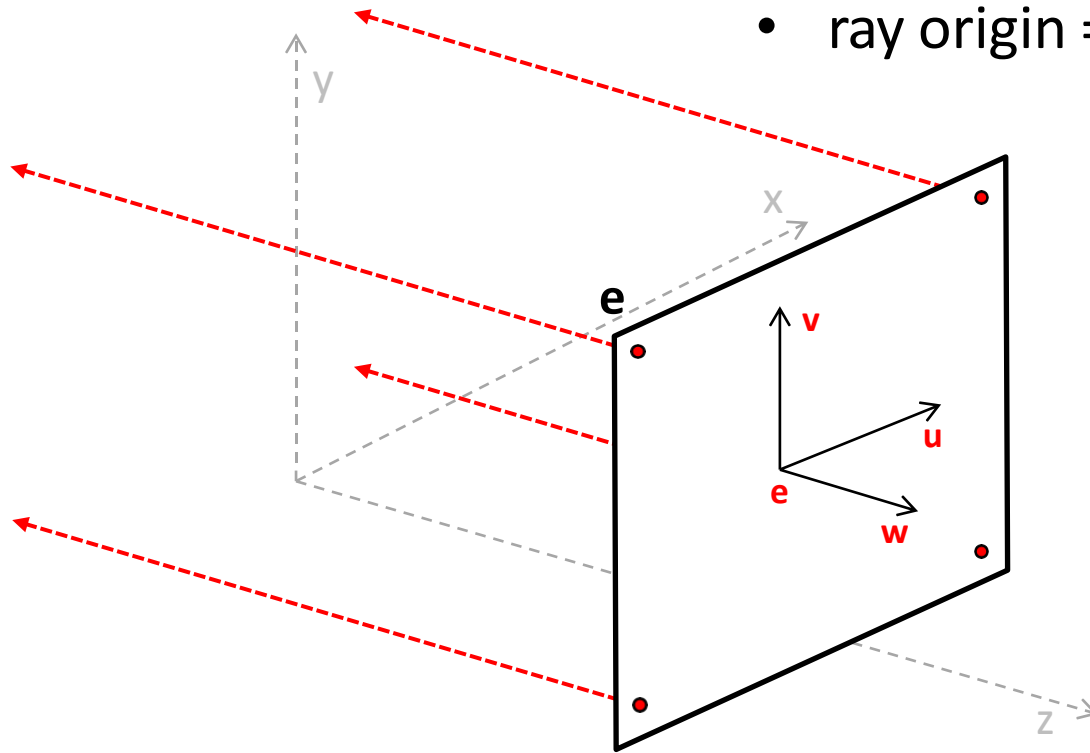


Image Plane (1/4)

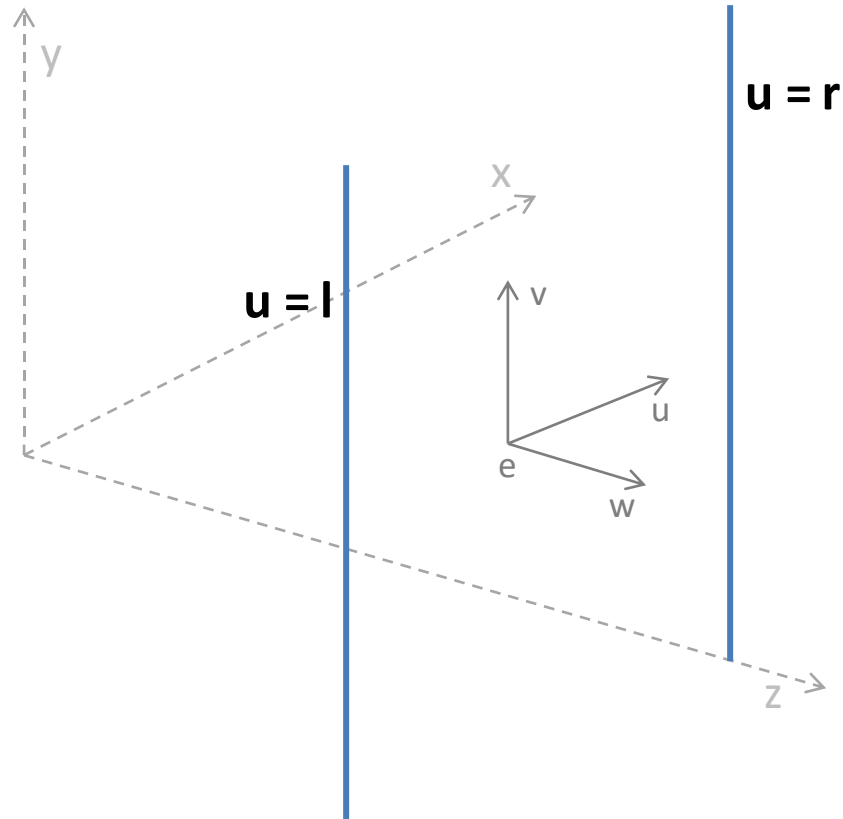


Image Plane (2/4)

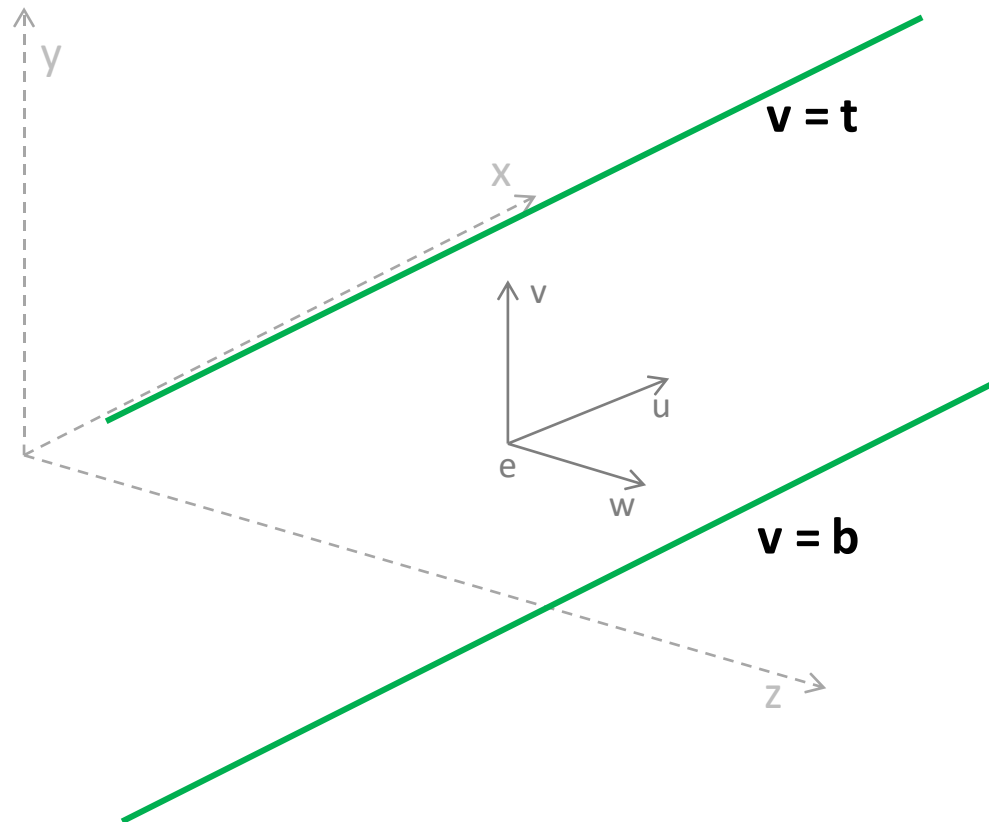


Image Plane (3/4)

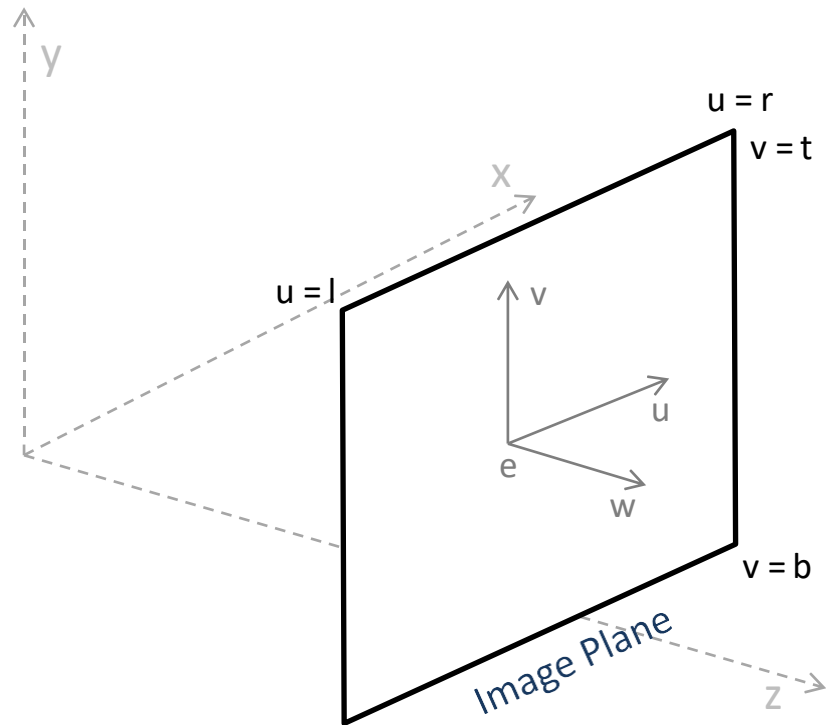
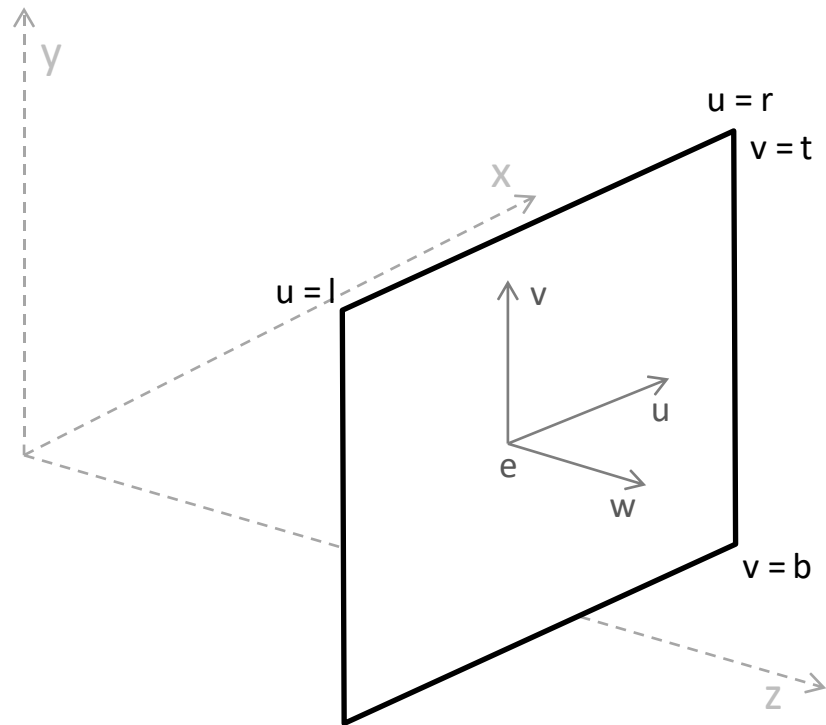
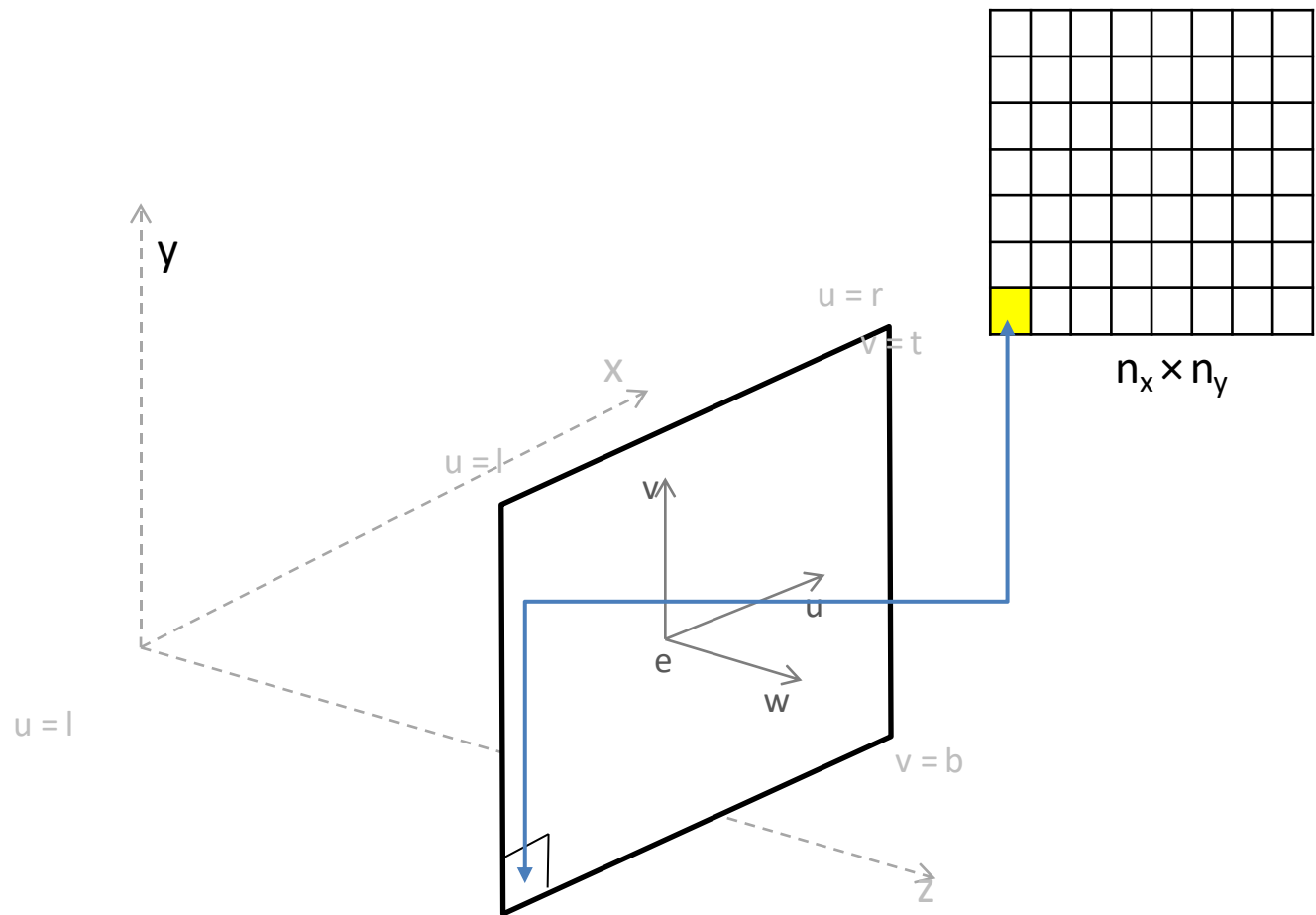


Image Plane (4/4)

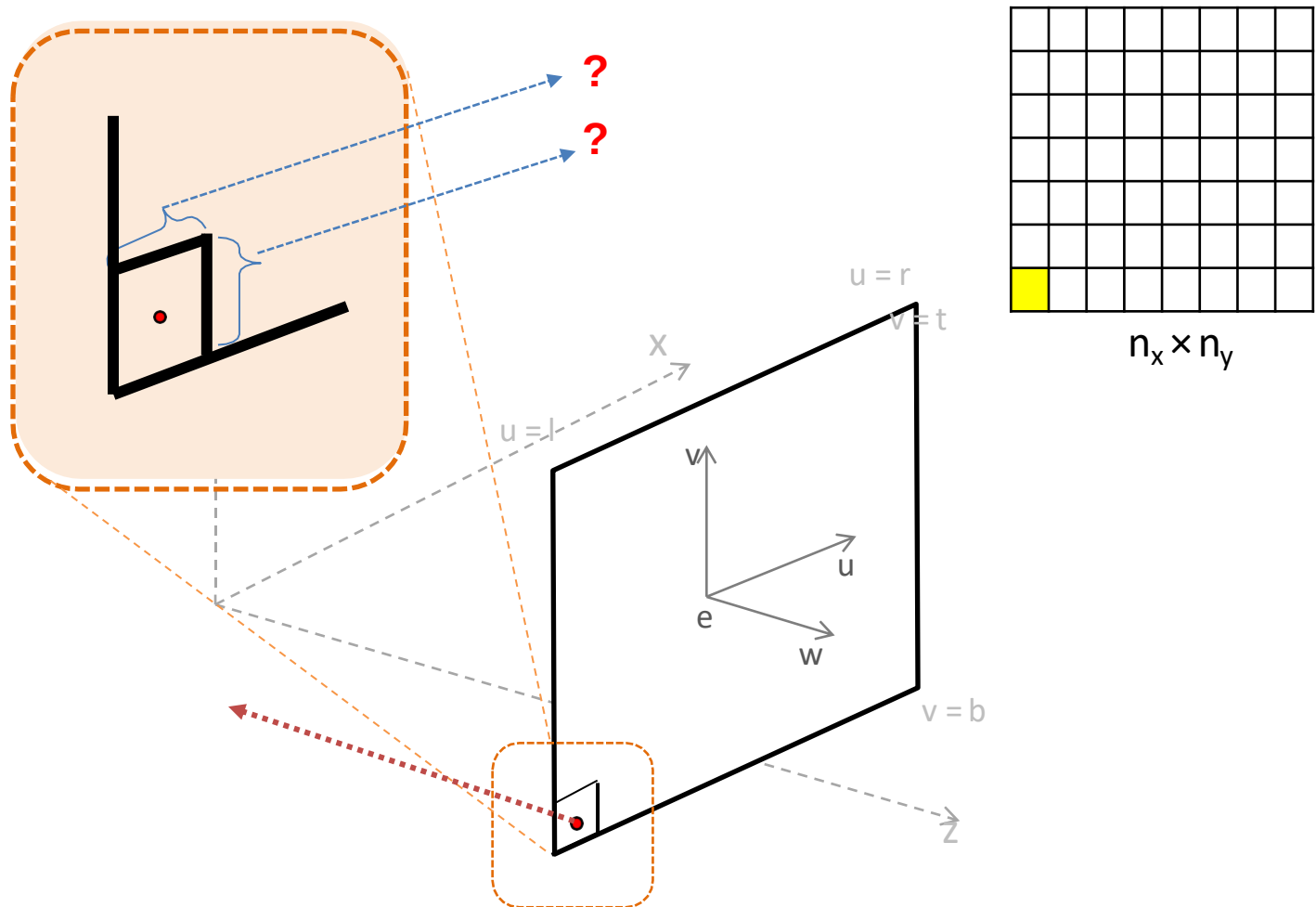
Q: determine the area of the image plane in terms of l , r , t and b .



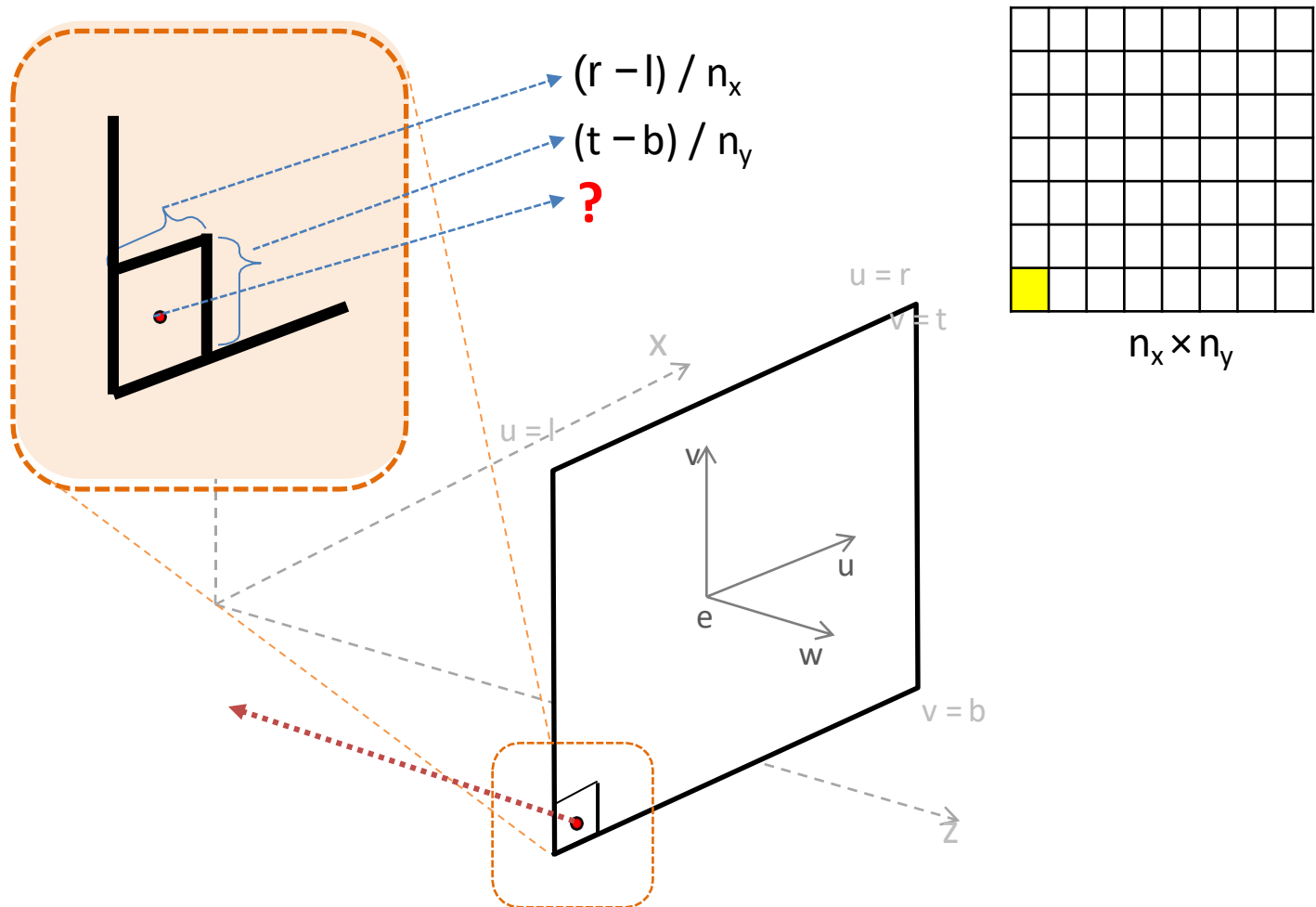
Raster Image $\leftrightarrow$ Image Plane (1/8)



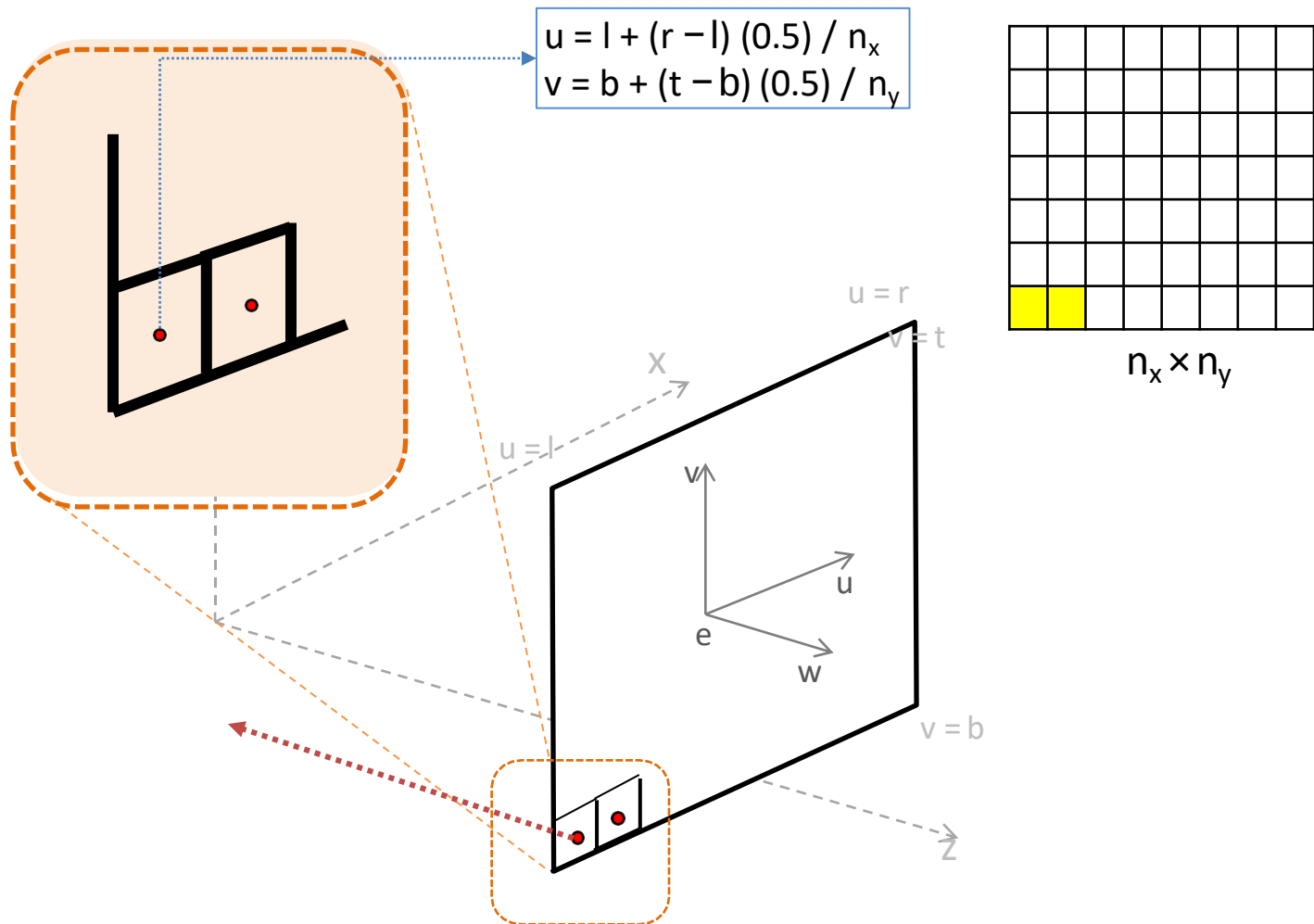
Raster Image $\leftrightarrow$ Image Plane (2/8)



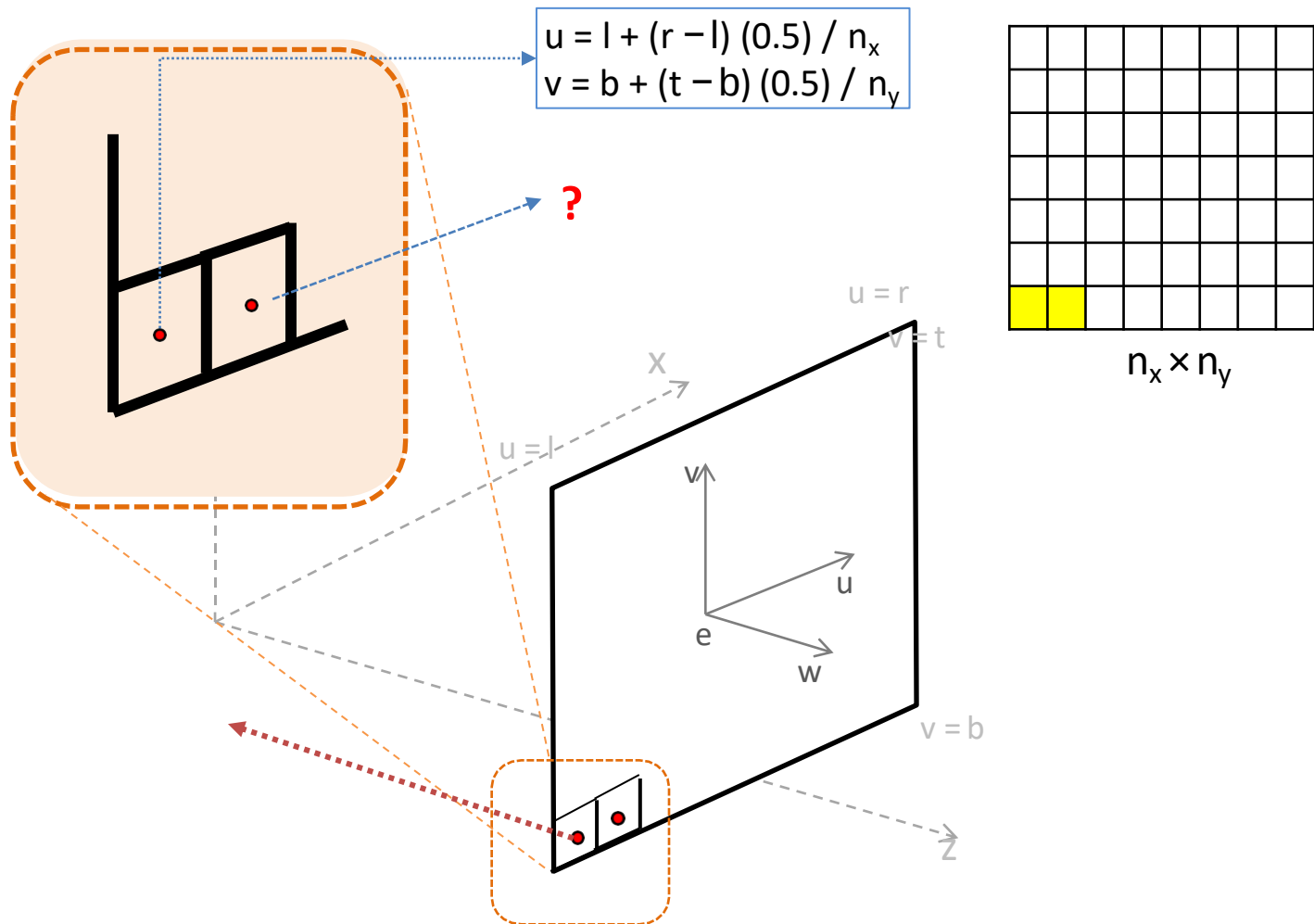
Raster Image $\leftrightarrow$ Image Plane (3/8)



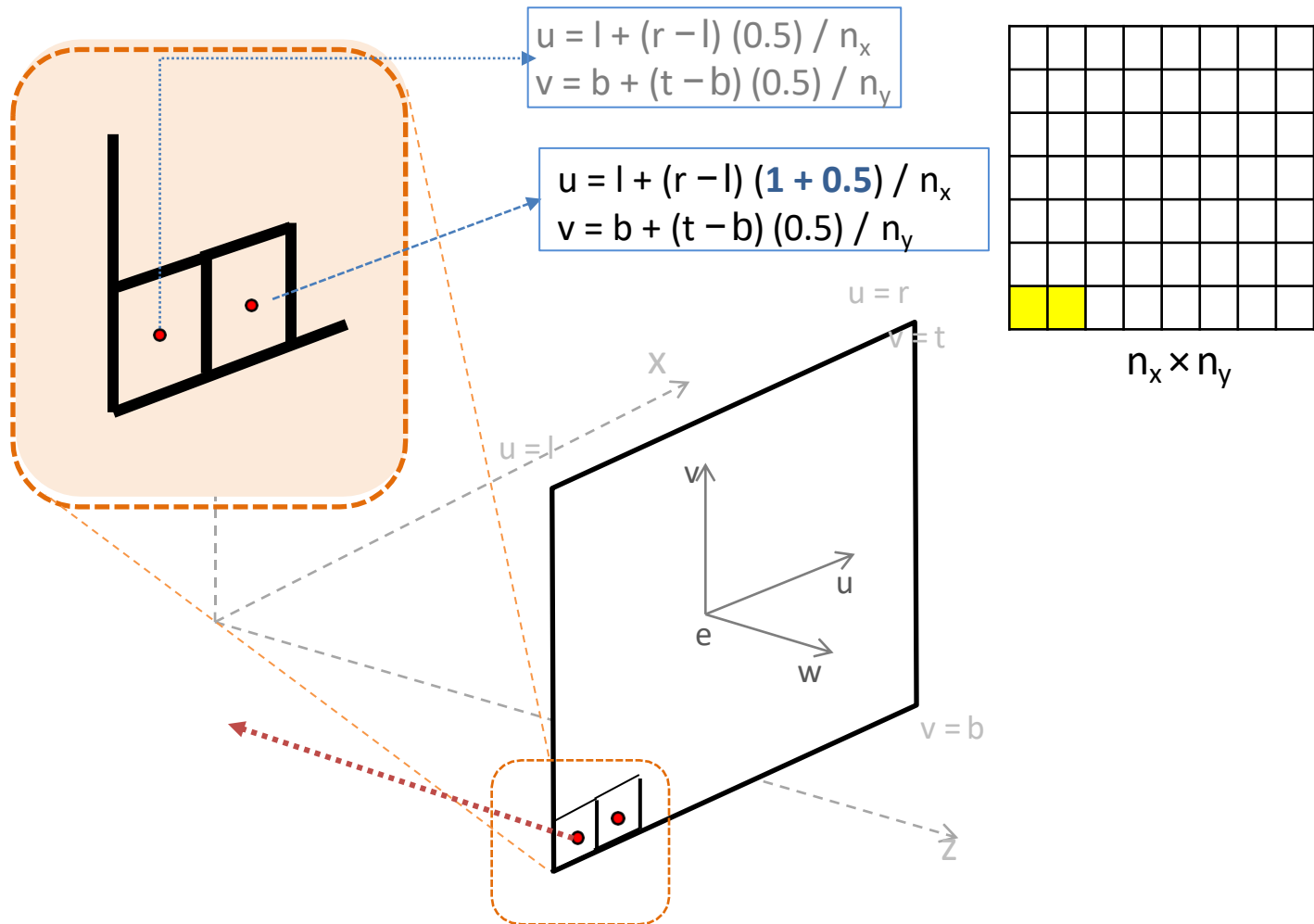
Raster Image $\leftrightarrow$ Image Plane (4/8)



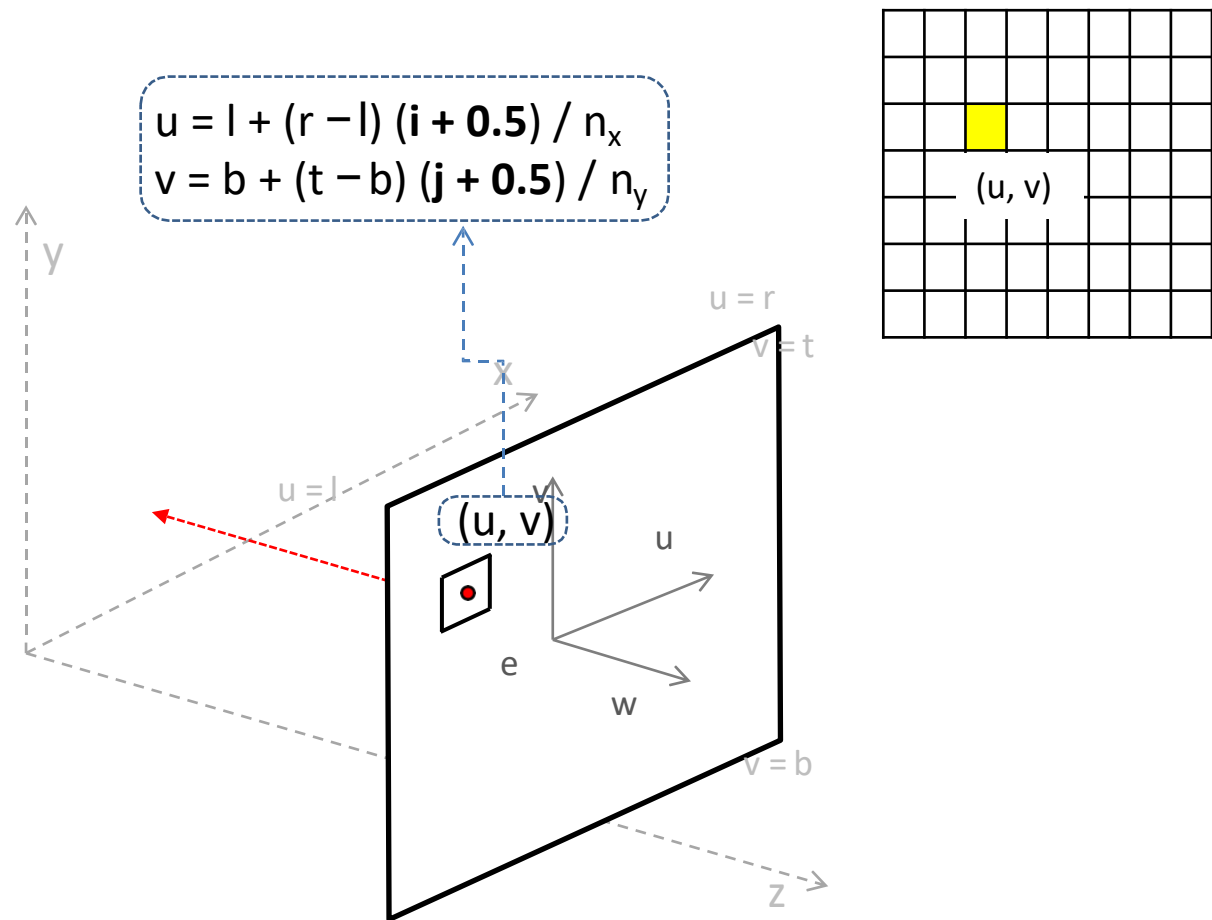
Raster Image $\leftrightarrow$ Image Plane (5/8)



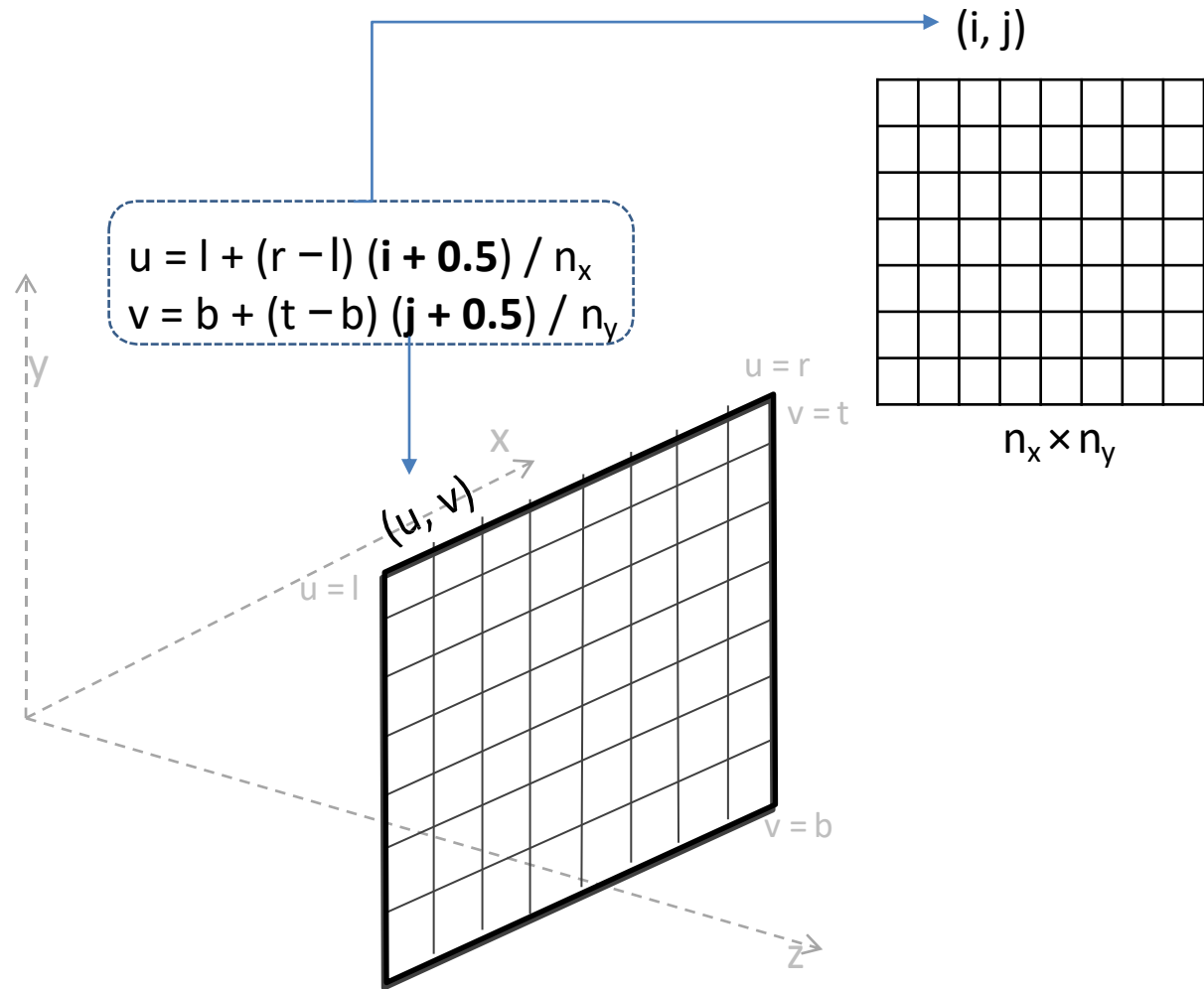
Raster Image $\leftrightarrow$ Image Plane (6/8)



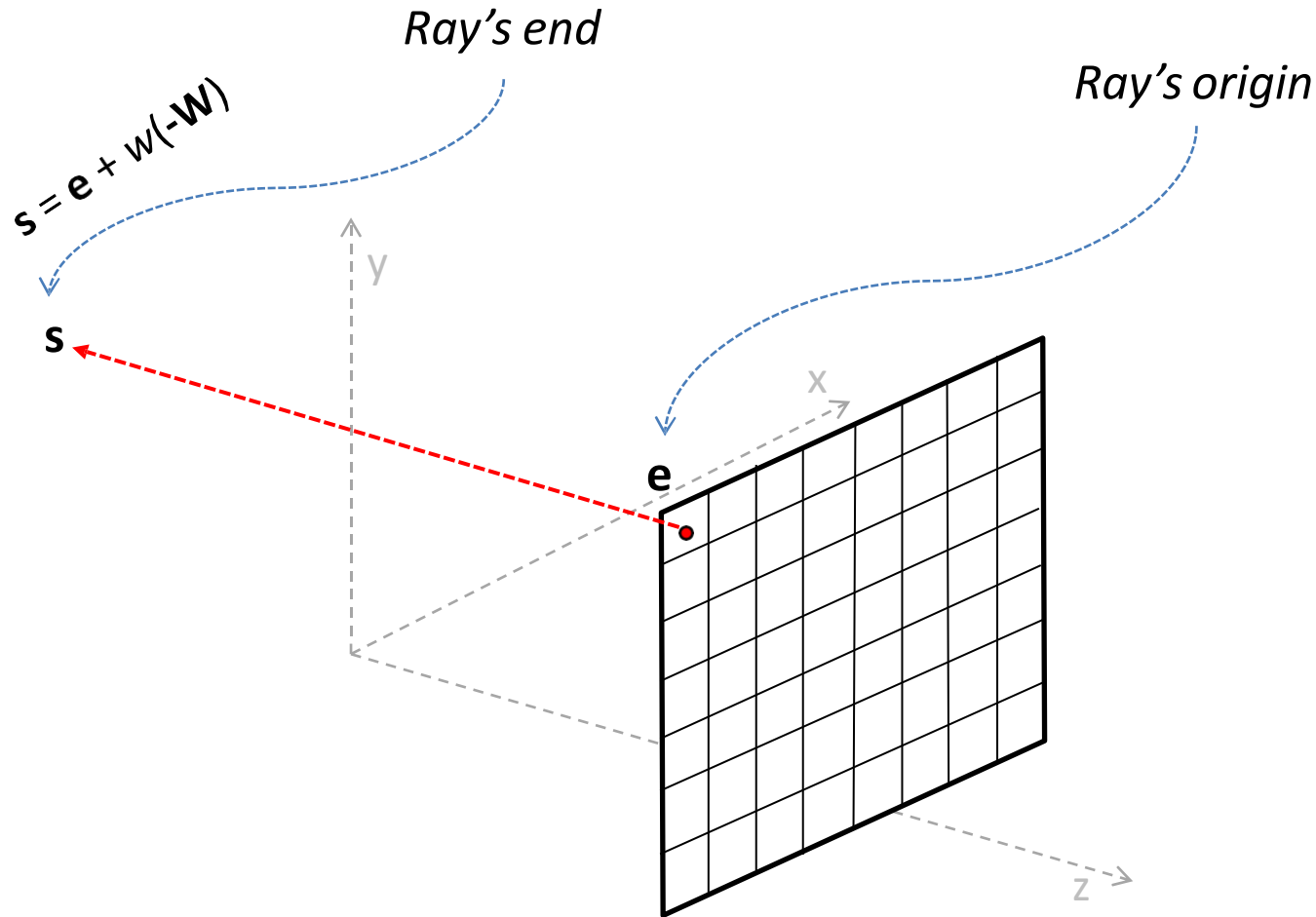
Raster Image $\leftrightarrow$ Image Plane (7/8)



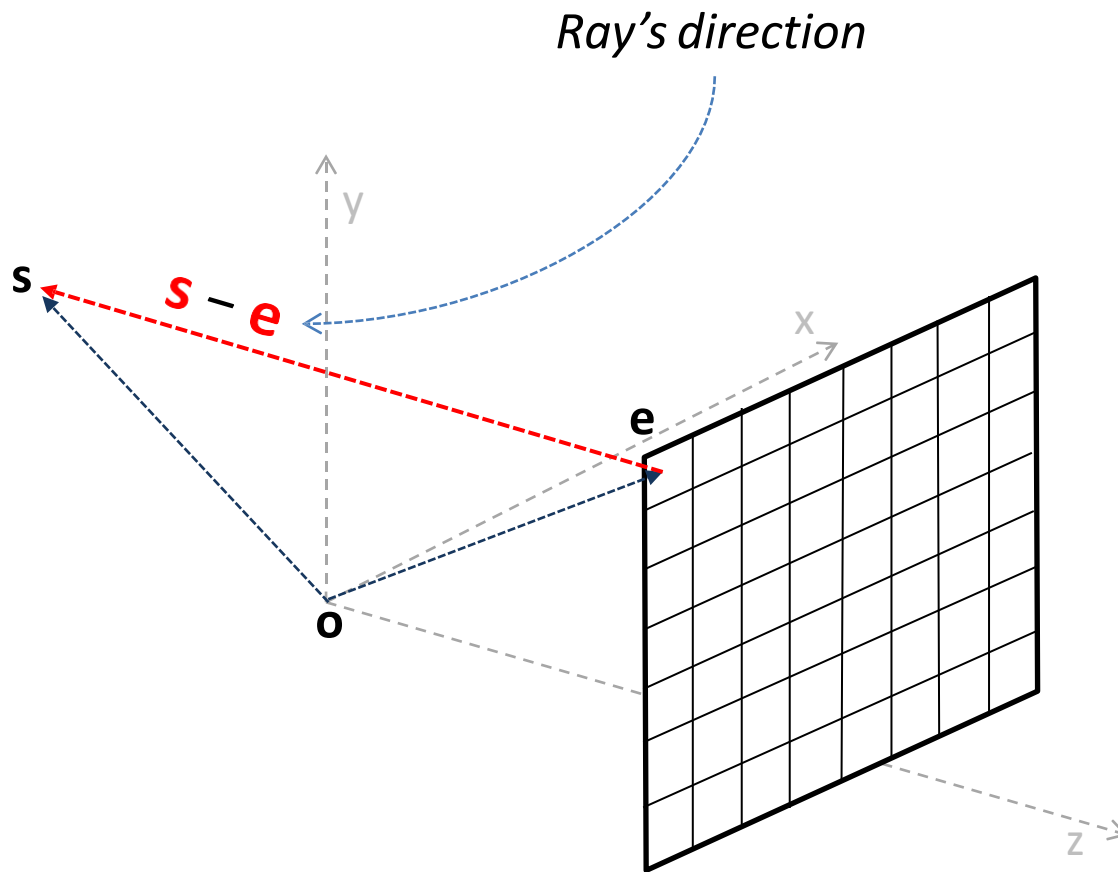
Raster Image $\leftrightarrow$ Image Plane (8/8)



Computing Viewing Rays (1/4)



Computing Viewing Rays (2/4)

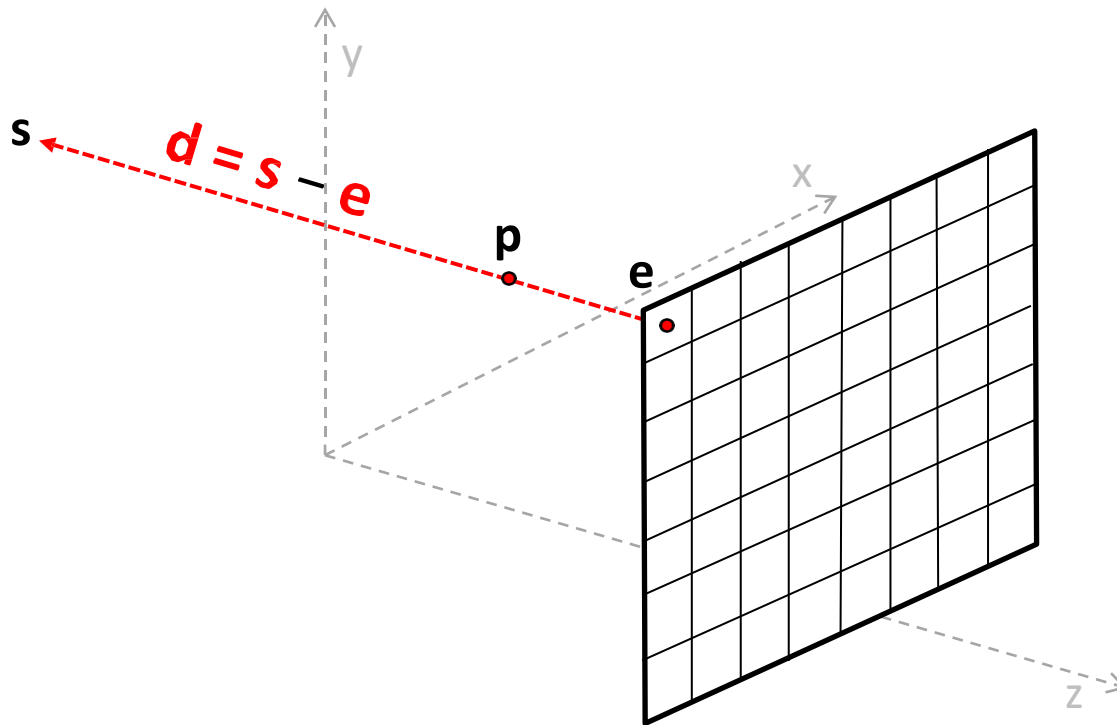


Computing Viewing Rays (3/4)

$$\begin{aligned}\mathbf{p} &= \mathbf{e} + t(\mathbf{s} - \mathbf{e}) \\ &= \mathbf{e} + t\mathbf{d}\end{aligned}$$

- Advancing from $\mathbf{e}$ along the vector $(\mathbf{s} - \mathbf{e})$

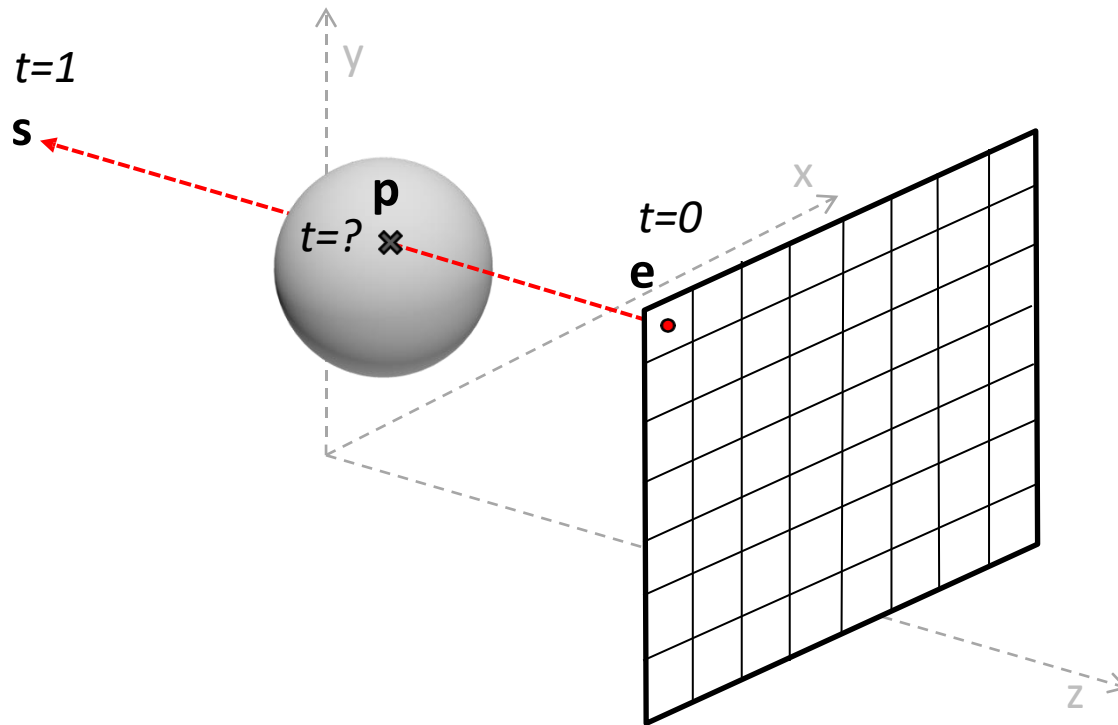
- With a fractional distance t to find the point $\mathbf{p}$



Computing Viewing Rays (4/4)

$$\begin{aligned}\mathbf{p} &= \mathbf{e} + t(\mathbf{s} - \mathbf{e}) \\ &= \mathbf{e} + t\mathbf{d}\end{aligned}$$

- We can use t to determine the intersection point $\mathbf{p}$

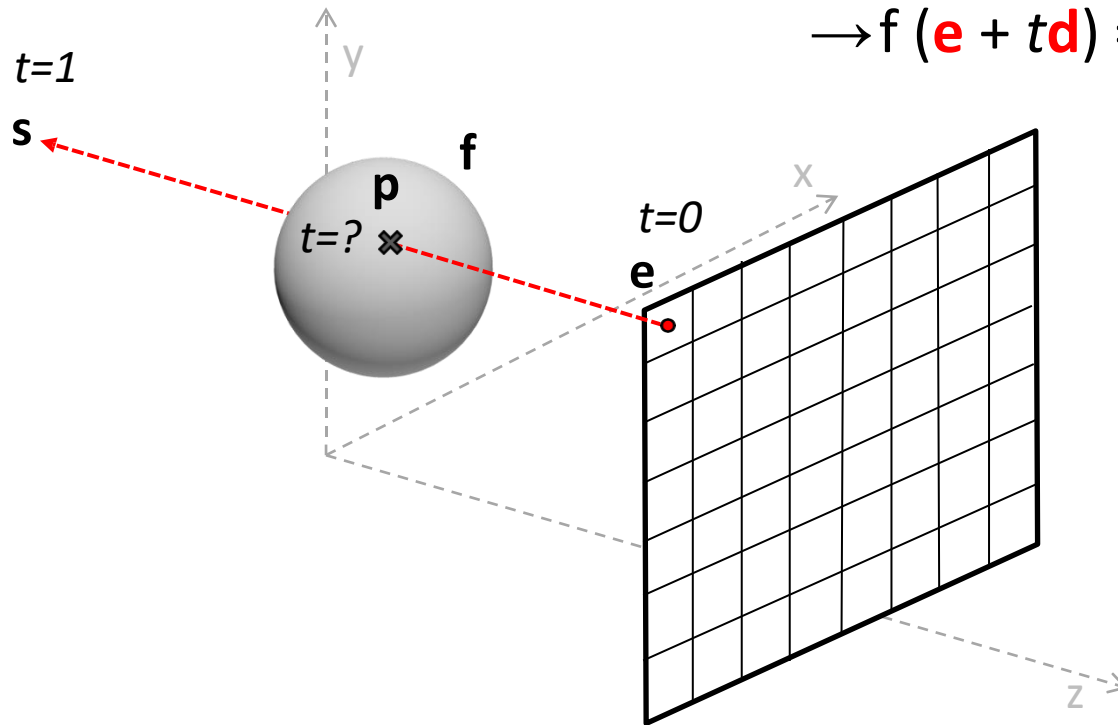


Ray - Sphere Intersection (1/8)

We have, $\mathbf{p} = \mathbf{e} + t(\mathbf{s} - \mathbf{e}) = \mathbf{e} + t\mathbf{d}$

$$\rightarrow f(\mathbf{p}) = 0$$

$$\rightarrow f(\mathbf{e} + t\mathbf{d}) = 0$$

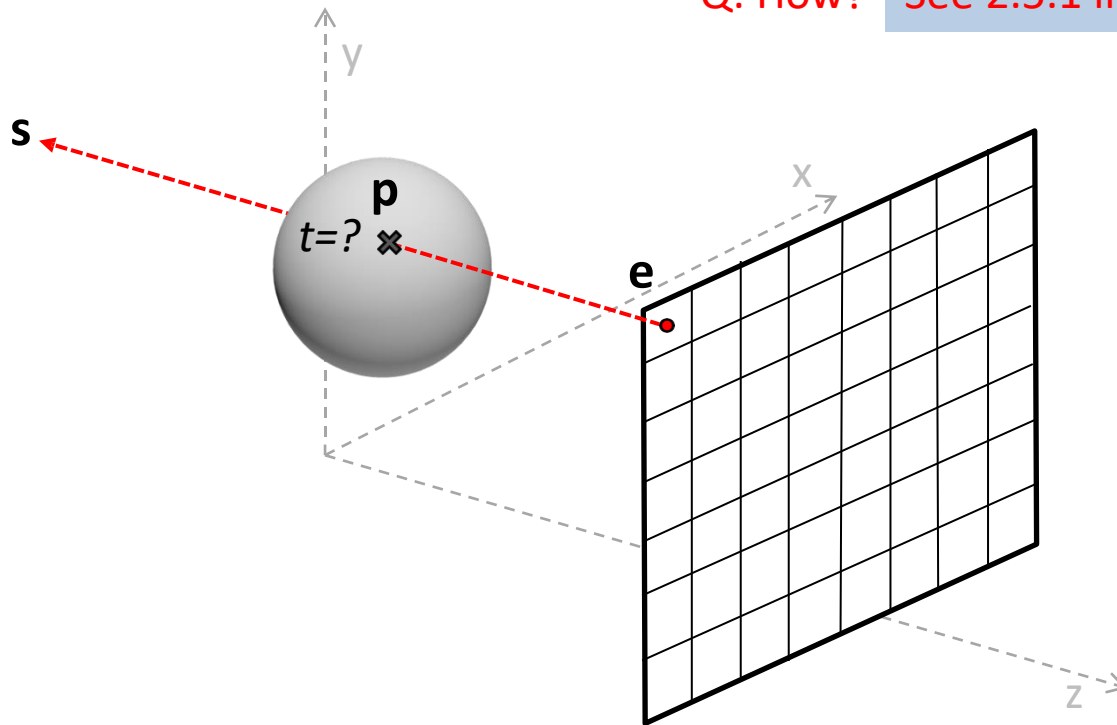


Ray - Sphere Intersection (2/8)

$$\rightarrow (x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - R^2 = 0$$

$$\rightarrow (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - R^2 = 0$$

Q: How? See 2.5.1 in Book



Ray - Sphere Intersection (3/8)

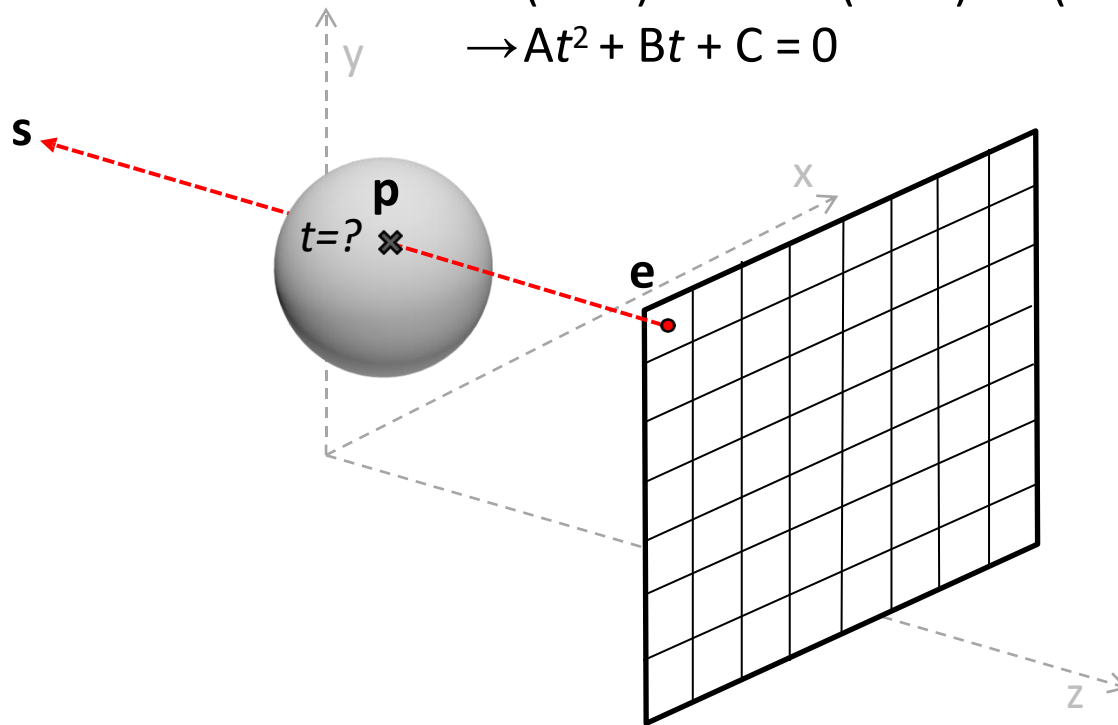
$$\rightarrow (x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - R^2 = 0$$

$$\rightarrow (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow (\mathbf{e} + t\mathbf{d} - \mathbf{c}) \cdot (\mathbf{e} + t\mathbf{d} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow (\mathbf{d} \cdot \mathbf{d}) t^2 + 2\mathbf{d} \cdot (\mathbf{e} - \mathbf{c}) t + (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow At^2 + Bt + C = 0$$



Ray - Sphere Intersection (4/8)

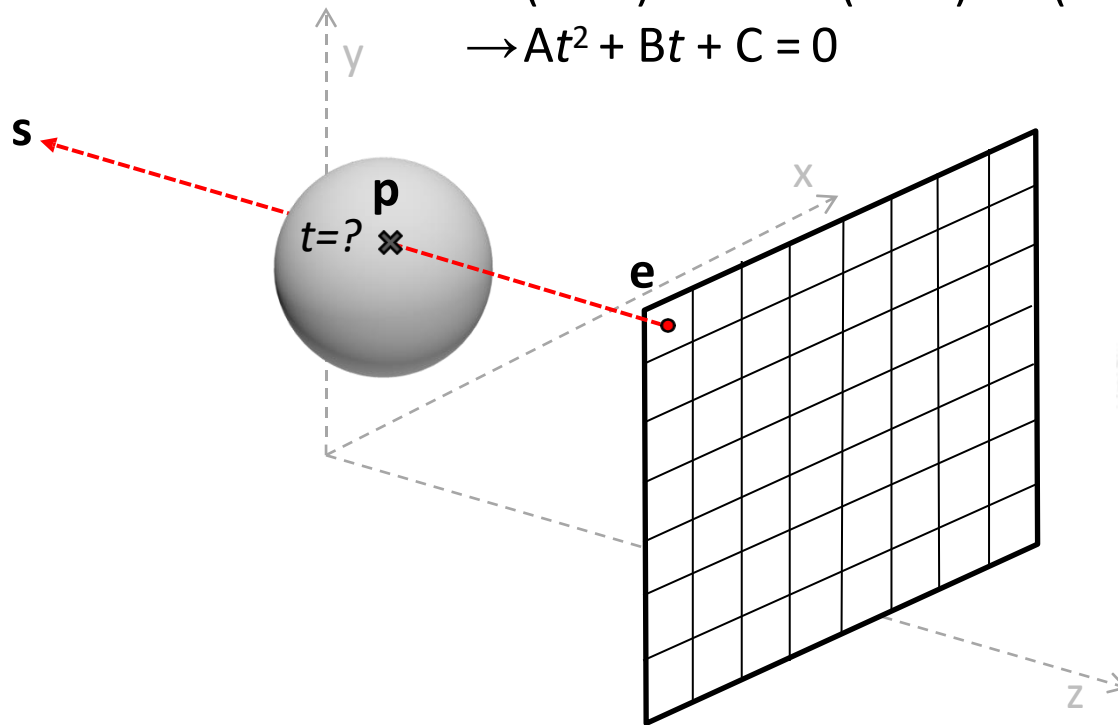
$$\rightarrow (x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - R^2 = 0$$

$$\rightarrow (\mathbf{p} - \mathbf{c}) \cdot (\mathbf{p} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow (\mathbf{e} + t\mathbf{d} - \mathbf{c}) \cdot (\mathbf{e} + t\mathbf{d} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow (\mathbf{d} \cdot \mathbf{d}) t^2 + 2\mathbf{d} \cdot (\mathbf{e} - \mathbf{c}) t + (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - R^2 = 0$$

$$\rightarrow At^2 + Bt + C = 0$$



$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

Ray - Sphere Intersection (5/8)

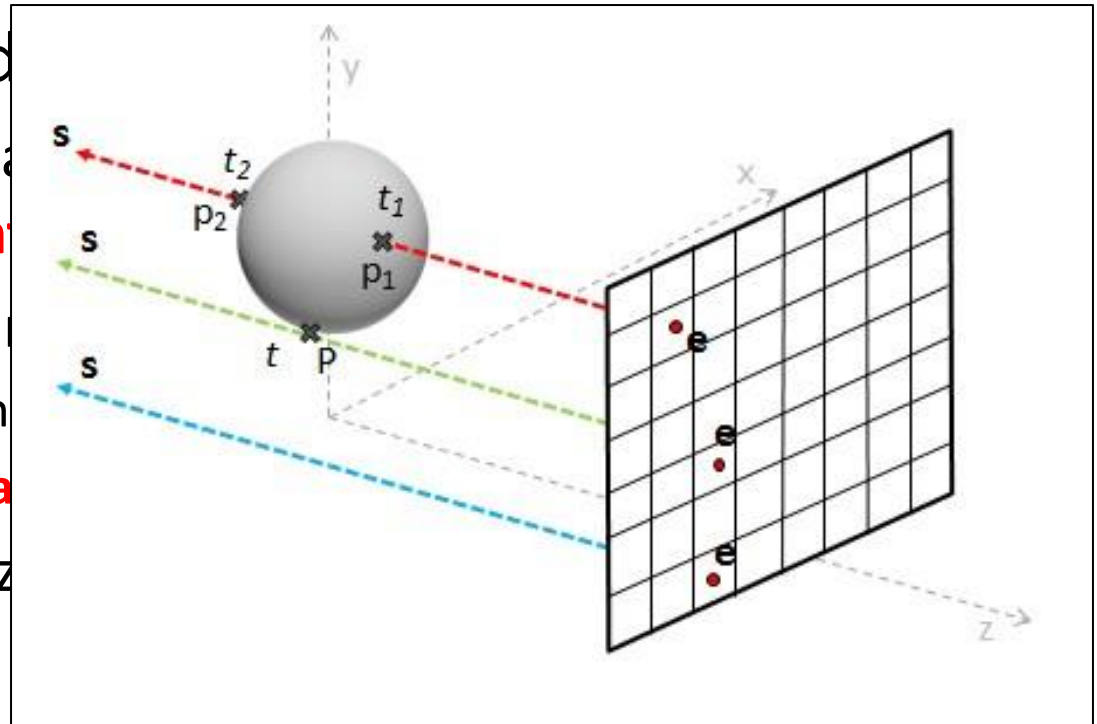
$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

- $B^2 - 4AC$, is called the discriminant and if it is –
 - **negative**: its square root is imaginary and the line and sphere **do not intersect**.
 - **positive**: there are two solutions –
 - one solution where the ray **enters** the sphere.
 - one where it **leaves**.
 - **zero**: the ray grazes the sphere, touching it at exactly **one point**.

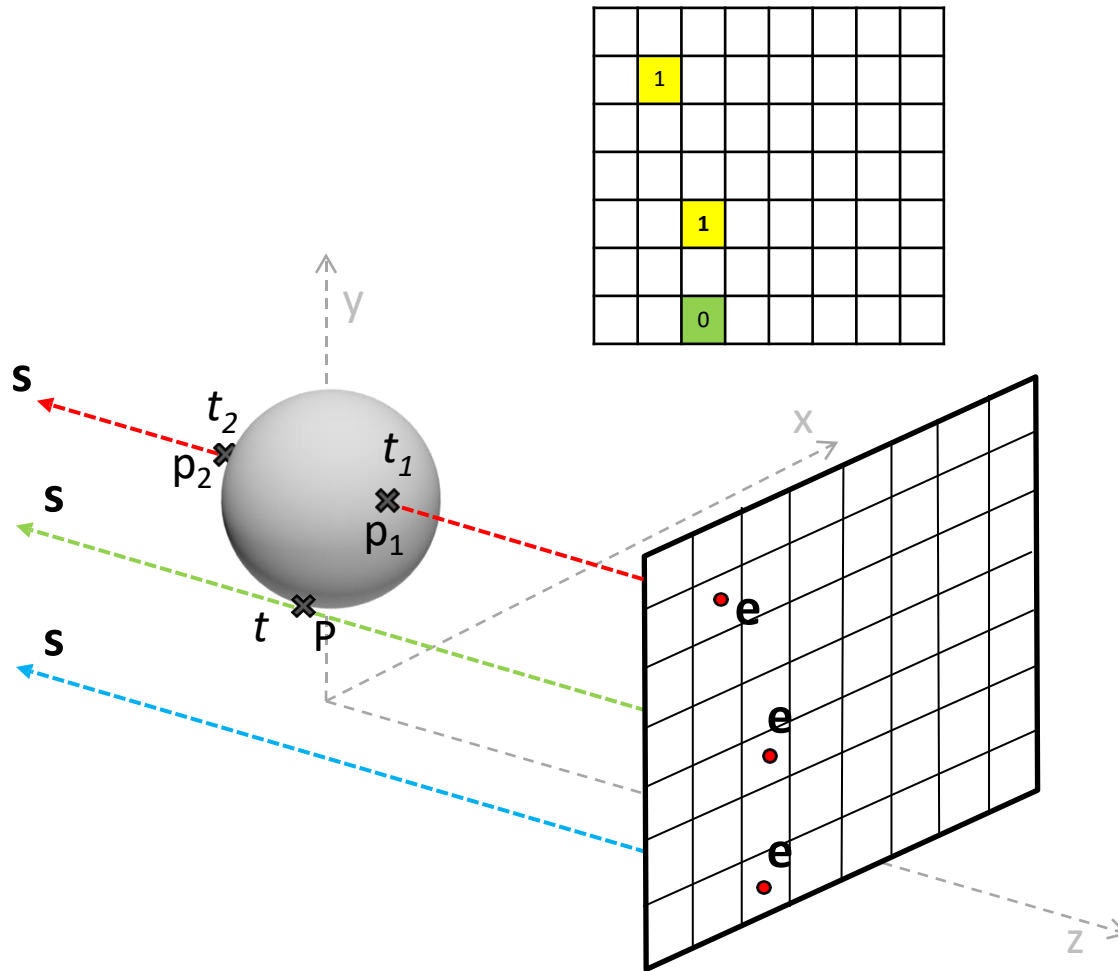
Ray - Sphere Intersection (5/8)

$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

- $B^2 - 4AC$, is called
 - **negative**: its square root is imaginary, sphere **do not intersect**
 - **positive**: there are two solutions
 - one solution where it **enters** the sphere
 - one where it **leaves** the sphere
 - **zero**: the ray grazes the sphere at a single **point**.



Ray - Sphere Intersection (6/8)



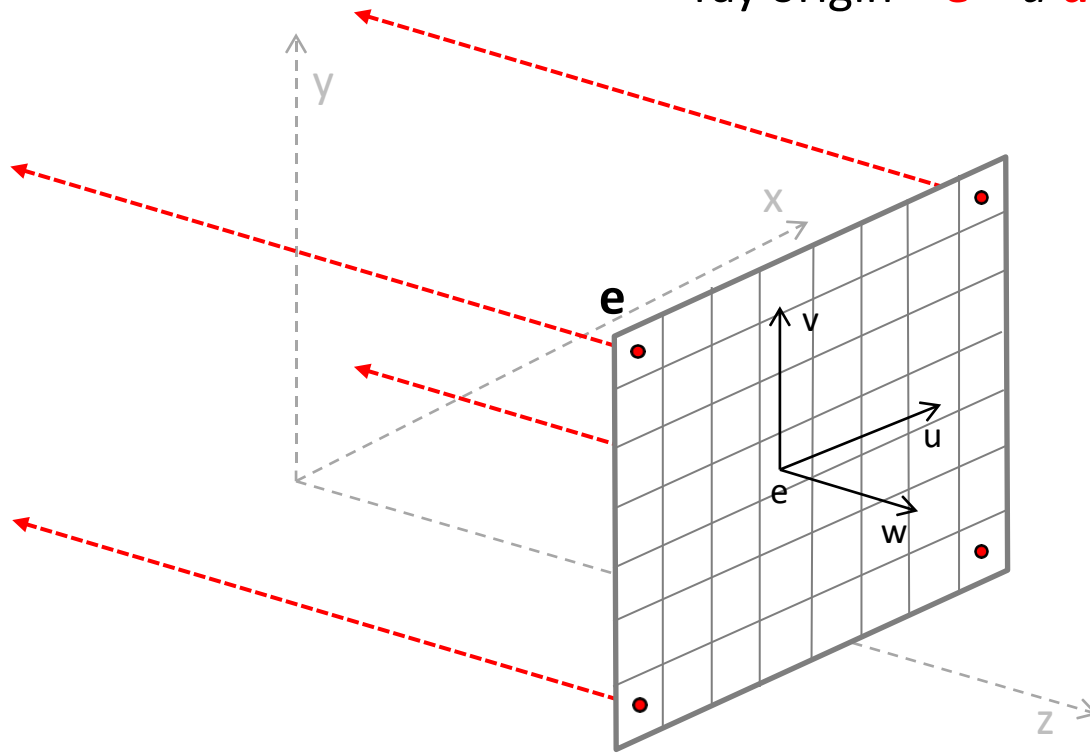
$$\begin{aligned} \mathbf{p} &= \mathbf{e} + t(\mathbf{s} - \mathbf{e}) \\ &= \mathbf{e} + t\mathbf{d} \end{aligned}$$

$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

Ray - Sphere Intersection (7/8)

Orthographic:

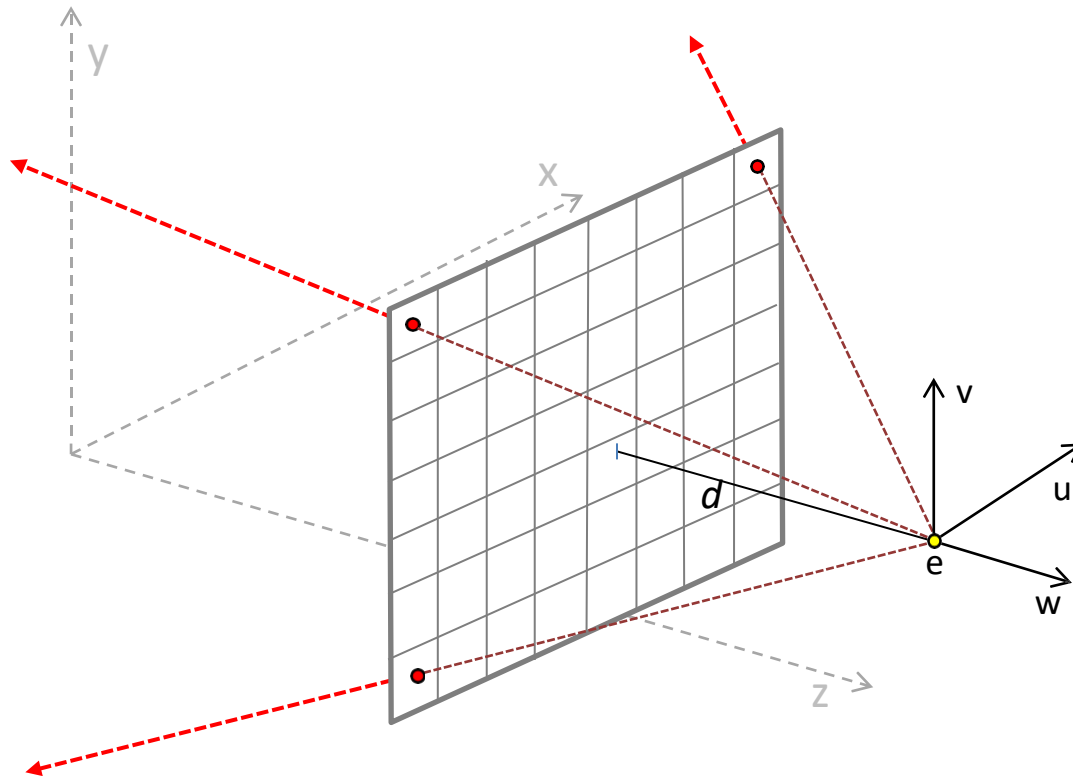
- ray direction = $-\mathbf{w}$
- ray origin = $\mathbf{e} + u \mathbf{u} + v \mathbf{v}$



Ray - Sphere Intersection (8/8)

Perspective:

- ray direction = $-d\mathbf{w} + u\mathbf{u} + v\mathbf{v}$
 - ray origin = $\mathbf{e}$
- Q: why?*

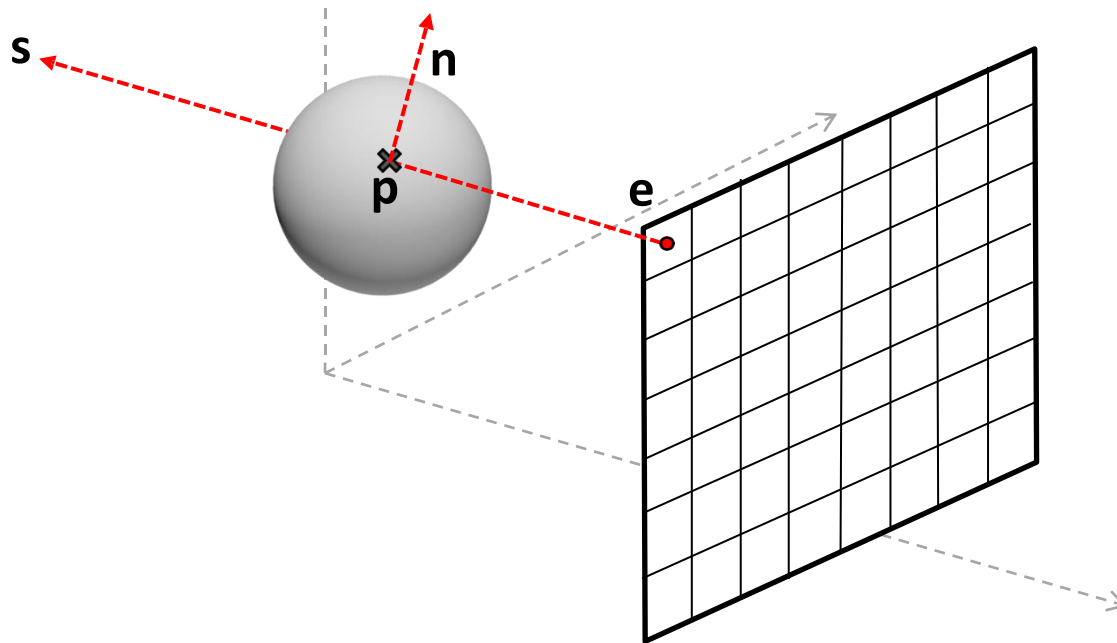


Shading (1/3)

Normal vector at point **p**:

- Gradient, $\mathbf{n} = 2 (\mathbf{p} - \mathbf{c})$.
- unit normal is $(\mathbf{p} - \mathbf{c})/R$.

[See section 2.5.4]

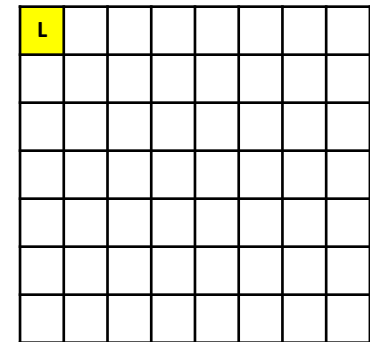
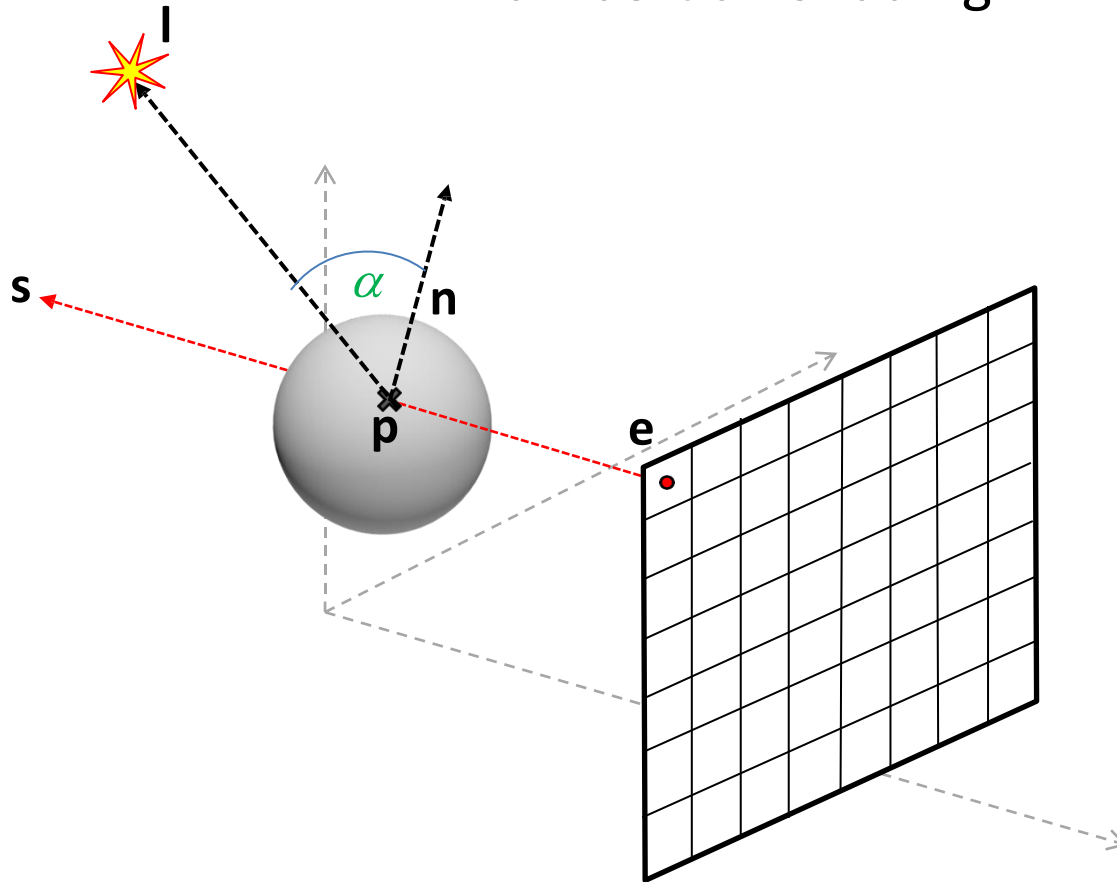


Shading (2/3)

Lambertian Shading: $L = k_d P \max(0, \mathbf{n} \cdot \mathbf{l})$

where,

- L = pixel color
- k_d = surface color
- P = intensity of the light source.



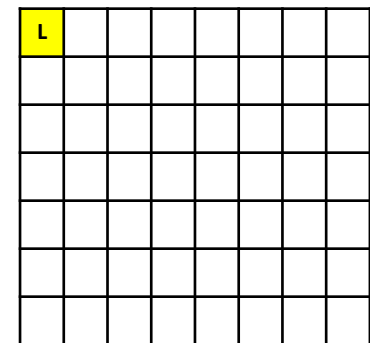
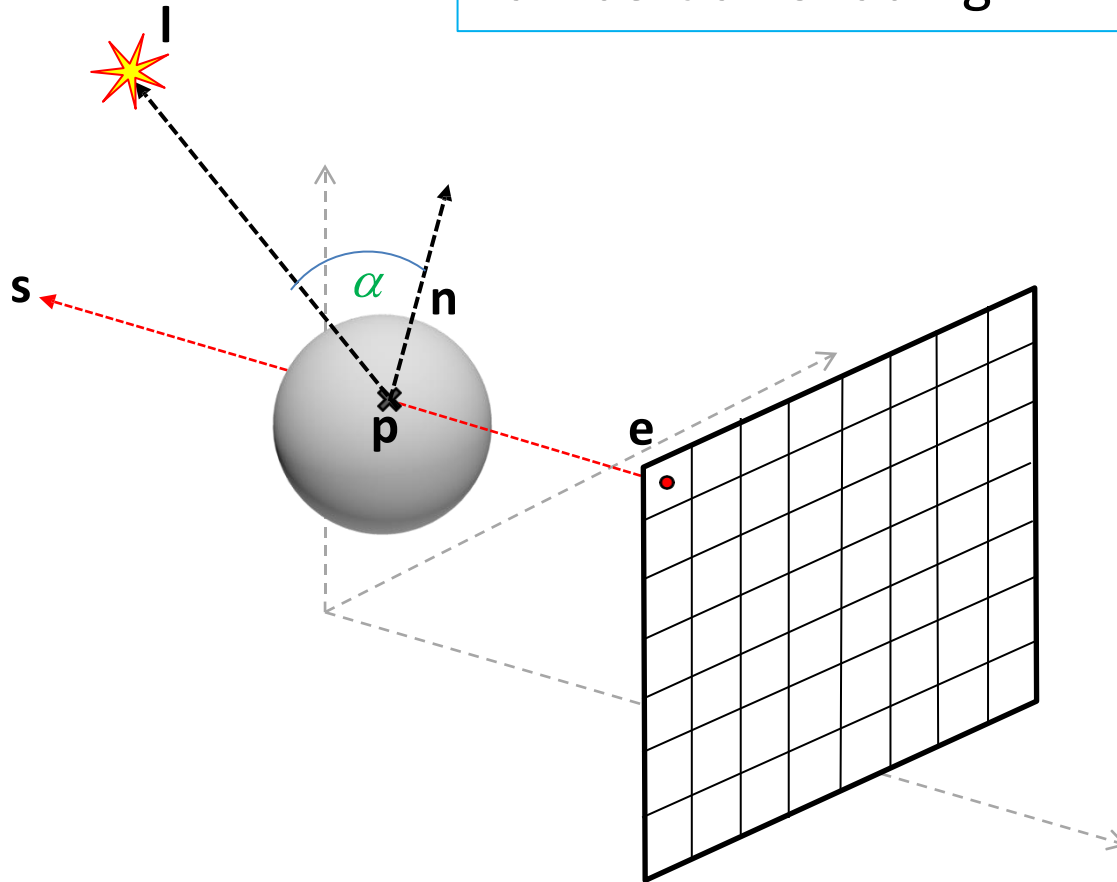
Shading (3/3)

Q: Are we considering angle in this formula? If yes – how?

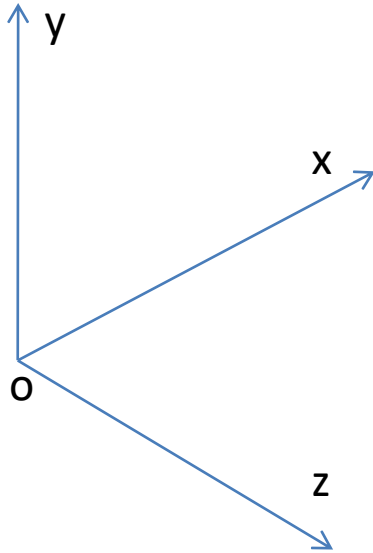
Lambertian Shading: $L = k_d P \max(0, \mathbf{n} \cdot \mathbf{l})$

where,

- L = pixel color
- k_d = surface color
- P = intensity of the light source.



Practice Problem (1/3)



Camera frame (*orthographic*):

- $\mathbf{e} = [4, 4, 6]$; $\mathbf{u} = [1, 0, 0]$; $\mathbf{v} = [0, 1, 0]$; $\mathbf{w} = [0, 0, 1]$
 - Plot the camera frame on the given axis.

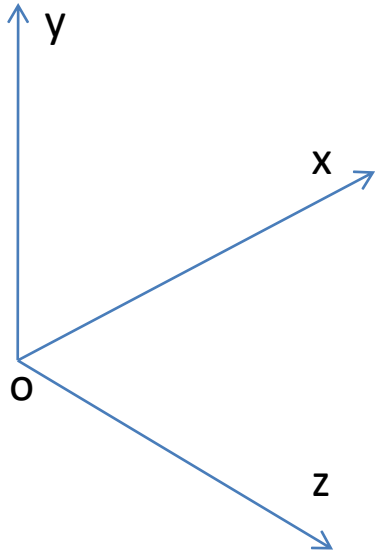
Viewing Ray:

- $\text{ray}_1.\text{origin} = \mathbf{e} + 2\mathbf{u} + 2\mathbf{v}$; $\text{ray}_1.\text{end} = [6, 6, 0]$
- $\text{ray}_2.\text{origin} = \mathbf{e} - 1\mathbf{u} + 1\mathbf{v}$; $\text{ray}_2.\text{end} = [4, 4, 0]$
 - Plot the origins for ray_1 and ray_2 .

Sphere: $f(x, y, z) = x^2 + y^2 + z^2 - (4)^2 = 0$

1. What are the intersecting points for ray_1 and ray_2 ?
2. Plot the intersecting points.

Practice Problem (2/3)



Camera frame (*orthographic*):

- $\mathbf{e} = [4, 4, 8]$; $\mathbf{u} = [1, 0, 0]$; $\mathbf{v} = [0, 1, 0]$; $\mathbf{w} = [0, 1, 0]$

Image Plane:

- left: $u = -5$; right: $u = 5$; top: $v = 4$; bottom: $v = -4$

1. Plot the image plane on the given axis.
2. For a 10×10 image matrix M , what is the position on the image plane for the ray origin at $M(4,3)$?
3. Will it intersect $f(x, y, z) = x^2 + y^2 + z^2 - 5^2 = 0$?

Practice Problem (3/3)

Consider the following parameters for an orthographic ray-tracing:

- *Camera frame:*

$$E = [-2, 7, 17]^T, U = [1, 0, 0]^T, V = [0, 1, 0]^T, W = [0, 0, 1]^T$$

- *Image plane:*

$$l = -15, r = 15, t = 10, b = -10$$

- *Raster image resolution:* 13×11
- *Sphere:* $(x+3)^2 + (y-5)^2 + (z-3)^2 = 64$

Determine the ray-sphere intersection point(s) for a ray (with *length* = 25) at the *center* of the raster image. Drawing figures is NOT mandatory.

Practice Problem (3/3)

Solution steps:

- Find u and v
$$u = l + (r - l)(i + 0.5) / nx$$
$$v = b + (t - b)(j + 0.5) / ny$$
- Determine the ray origin, $\mathbf{e} = \mathbf{E} + u\mathbf{U} + v\mathbf{V}$
- Find ray end point, $\mathbf{s} = \mathbf{e} + w(-\mathbf{W})$
- Determine, $\mathbf{d} = \mathbf{s} - \mathbf{e}$
- Determine, $D = B^2 - 4AC$
$$A = \mathbf{d} \cdot \mathbf{d}$$
$$B = 2\mathbf{d} \cdot (\mathbf{e} - \mathbf{c})$$
$$C = (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - R^2$$
- Determine the intersection parameter, t
$$t_1 = (-B + \sqrt{D}) / (2A)$$
$$t_2 = (-B - \sqrt{D}) / (2A)$$
- Determine the intersection point,
$$\mathbf{P}_1 = \mathbf{e} + t_1(\mathbf{s} - \mathbf{e})$$
$$\mathbf{P}_2 = \mathbf{e} + t_2(\mathbf{s} - \mathbf{e})$$

Further Reading

- Fundamentals of Computer Graphics, 4th Edition -
Chapter 4